

MALIN HEAD, Ireland

WMO: 039800

Lat: 55.367N

Lon: 7.333W

Elev: 82

StdP: 14.65

Time Zone: 1.00 (IRL)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	32.4	34.5	24.4	18.6	36.0	26.8	20.8	37.6	48.8	42.8	45.3	43.9	11.9	190	0.837

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.5	67.8	61.5	65.4	60.3	63.4	59.2	62.8	66.2	61.4	64.3	60.1	62.7	15.7	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.2	81.1	64.6	59.9	77.4	63.1	58.7	74.0	61.7	28.4	66.4	27.4	64.5	26.5	62.8	68.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 41.3	36.4	32.5	DB	28.9	73.8	2.2	2.4	27.3	75.6	26.0	77.0	24.7	78.3	23.1	80.1	(4)
(5)			WB	27.3	65.9	2.0	1.3	25.9	66.8	24.7	67.6	23.5	68.3	22.0	69.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	50.1	43.1	43.1	44.5	47.4	50.9	54.9	58.0	58.5	56.3	52.1	47.4	44.2	(6)
(7)		DBStd	6.72	3.98	4.00	4.02	3.99	3.79	3.34	3.14	2.72	3.11	3.63	3.86	4.38	(7)
(8)		HDD50	1018	216	196	175	98	32	2	0	0	0	20	94	185	(8)
(9)		HDD65	5456	680	614	635	529	436	304	220	203	261	400	528	646	(9)
(10)		CDD50	1043	1	2	5	19	61	148	247	264	190	85	16	5	(10)
(11)		CDD65	3	0	0	0	0	0	0	2	1	0	0	0	0	(11)
(12)		CDH74	3	0	0	0	0	1	0	1	1	0	0	0	0	(12)
(13)		CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)
(14)	Wind	WSAvg	16.8	20.2	19.6	18.4	16.2	14.9	13.9	12.7	13.9	15.8	17.9	18.8	19.5	(14)
(15)	Precipitation	PrecAvg	43.1	4.6	3.3	3.3	2.5	2.4	2.8	3.1	3.7	3.8	4.5	4.6	4.5	(15)
(16)		PrecMax	55.2	8.0	7.9	6.0	4.9	5.4	5.6	6.0	7.2	8.5	9.9	9.0	10.7	(16)
(17)		PrecMin	33.1	0.9	0.4	0.7	0.7	0.6	1.1	0.6	0.5	0.9	1.2	1.0	1.7	(17)
(18)		PrecStd	4.8	1.8	1.7	1.3	1.0	1.1	0.9	1.2	1.6	1.6	1.9	1.6	1.9	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.3	55.2	57.1	62.4	66.3	68.9	71.3	71.3	68.3	63.2	58.4	56.5	(19)
(20)			MCWB	51.5	51.5	51.6	53.9	57.4	60.1	63.3	64.2	61.5	58.7	55.4	53.5	(20)
(21)		2%	DB	52.0	52.7	53.6	57.8	61.4	64.5	67.5	67.1	65.0	60.7	56.0	53.9	(21)
(22)			MCWB	49.8	49.8	49.8	52.2	55.2	58.6	61.8	61.9	60.3	57.3	53.6	51.5	(22)
(23)		5%	DB	50.4	50.9	51.7	55.0	58.5	61.8	64.7	64.6	62.7	58.9	54.4	52.2	(23)
(24)			MCWB	48.5	48.5	48.5	50.7	53.7	57.2	60.6	60.5	58.8	55.8	52.1	49.9	(24)
(25)		10%	DB	49.1	49.2	50.2	52.9	56.3	59.8	62.7	62.7	60.9	57.4	53.0	50.6	(25)
(26)			MCWB	47.1	47.0	47.5	49.1	52.3	56.1	59.0	59.3	57.5	54.6	50.7	48.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.1	52.6	52.7	55.9	59.3	62.0	64.8	65.1	63.2	59.6	56.2	54.3	(27)
(28)			MCD B	53.7	54.2	55.5	60.5	64.3	66.1	69.2	69.2	66.6	62.2	57.8	56.2	(28)
(29)		2%	WB	50.3	50.4	50.6	53.0	56.3	59.4	62.6	62.9	61.1	57.9	54.2	52.0	(29)
(30)			MCD B	51.7	52.1	52.7	56.8	59.9	63.3	66.2	66.0	63.9	60.1	55.6	53.7	(30)
(31)		5%	WB	49.0	49.0	49.3	51.1	54.4	57.9	61.1	61.4	59.7	56.6	52.7	50.4	(31)
(32)			MCD B	50.4	50.8	51.2	54.4	57.8	61.1	64.2	63.9	62.1	58.6	54.2	52.1	(32)
(33)		10%	WB	47.5	47.3	48.0	49.6	53.0	56.6	59.6	60.0	58.3	55.1	51.2	48.7	(33)
(34)			MCD B	49.0	49.2	50.0	52.3	56.1	59.4	62.4	62.2	60.6	57.2	52.9	50.4	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	6.8	7.2	7.7	8.3	8.4	7.7	7.7	7.5	7.1	6.7	6.9	(35)	
(36)			MCDBR	7.7	7.7	9.3	11.6	12.0	11.3	11.2	10.8	10.3	8.6	7.4	8.1	(36)
(37)			MCWBR	7.8	7.1	7.2	7.7	7.8	7.4	7.4	7.4	7.4	7.2	7.3	8.2	(37)
(38)		5% WB	MCDBR	7.5	7.7	8.4	10.5	10.9	10.2	10.3	9.3	8.9	8.2	7.3	8.1	(38)
(39)			MCWBR	7.8	7.5	7.1	7.6	7.9	7.4	7.3	7.2	7.4	7.6	7.7	8.3	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.345	0.353	0.372	0.382	0.388	0.376	0.379	0.365	0.363	0.349	0.335	0.331	(40)	
(41)		taud	2.202	2.289	2.303	2.324	2.355	2.425	2.423	2.476	2.466	2.410	2.271	2.173	(41)	
(42)		Ebn at Noon	173	225	251	266	271	276	272	270	253	225	177	147	(42)	
(43)		Edh at Noon	19	25	31	35	36	34	34	30	27	22	18	16	(43)	
(44)	All-Sky Solar Radiation	RadAvg	159	377	722	1209	1559	1587	1473	1191	811	447	198	114	(44)	
(45)		RadStd	17	33	57	83	135	151	101	96	46	40	18	11	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	+1.44	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (28 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page